A dynamical singlet moment in complex DSSYK Gaussian contractions and a symmetry reduction

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Abstract

The static-patch programme needs singlet dynamics that can be compared with bulk propagation. We organize the first derivative singlet's equal-time strength and second frequency moment as exact Gaussian contractions. Applying the established multiplicity-free decomposition of fermionic operator spaces eliminates the linear Gaussian component of its time derivative. The dynamical moment consequently reduces to a positive cubic tensor norm. A complete finite-system contraction supplies a reproducible benchmark for the subsequent double-scaled calculation.

1 The physical quantity to calculate

Our target remains the singlet-to-bulk-particle step of the Susskind programme [1]. In the conventions of the preceding dictionary note,

$$H = \sum_{I,J} G_{IJ} c_I^\dagger c_J, \quad \mathbb{E} G_{IJ} G_{KL} = w \delta_{IL} \delta_{JK}, \quad w = 4^p \frac{\mathcal{J}_s^2}{\lambda} \left(\frac{p!}{N^p} \right)^2, \quad \lambda = \frac{p^2}{N}. \tag{1}$$

Here G is Hermitian, each index set has size p, the CAR is standard, and $H = H_s$ is in string units. At infinite temperature $\mathrm{tr} = 2^{-N}\,\mathrm{Tr}$. Define

$$A = \sum_i [H, c_i]^\dagger [H, c_i], \qquad K = [H, A]. \tag{2}$$

With $C(t) = \mathbb{E}\,\mathrm{tr}(A(t)A)$ and $S(\omega) = \int dt\, e^{i\omega t} C(t)$,

$$\mu_0 = \mathbb{E}\,\mathrm{tr}\, A^2, \qquad \boxed{\mu_2 = \int \frac{d\omega}{2\pi} \omega^2 S(\omega) = \mathbb{E}\,\mathrm{tr}\, K^\dagger K = -C''(0).} \tag{3}$$

The ω^2 weight removes every zero-frequency contribution. Thus μ_2 is a direct measure of singlet dynamics even when the equal-time strength is dominated by conserved components. The next physical target is its double-scaled size, followed by enough further moments to resolve the frequency dependence. A mass assignment will then require the bulk propagation kernel, rather than identifying a frequency moment with a mass.

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2 A contraction formula that keeps the same couplings

Choose real standard Gaussian coordinates x_a for the GUE and absorb their variances into Hermitian matrices V_a, so $H = \sum_a x_a V_a$ and $\mathbb{E} x_a x_b = \delta_{ab}$. The same coordinates occur in A:

$$D_{ai} = [V_a, c_i], \tag{4}$$

$$B_{ab} = \frac{1}{2}\sum_i (D^\dagger_{ai} D_{bi} + D^\dagger_{bi} D_{ai}), \qquad A = \sum_{ab} x_a x_b B_{ab}, \tag{5}$$

$$C_{abc} = \frac{1}{3}([V_a, B_{bc}] + [V_b, B_{ac}] + [V_c, B_{ab}]), \qquad K = \sum_{abc} x_a x_b x_c C_{abc}. \tag{6}$$

B is symmetric and Hermitian; C is fully symmetric and anti-Hermitian. Write $\|X\|^2 = \operatorname{tr} X^\dagger X$, and set

$$\overline{A} = \sum_a B_{aa}, \qquad L_c = \sum_a C_{aac}. \tag{7}$$

Ordinary Gaussian Wick contraction, with no large-N approximation, gives

$$\mu_0 = \|\overline{A}\|^2 + 2\sum_{ab} \|B_{ab}\|^2, \tag{8}$$

$$\mu_2 = 6\sum_{abc} \|C_{abc}\|^2 + 9\sum_c \|L_c\|^2. \tag{9}$$

The factors 6 and 9 count, respectively, the pairings with three cross-contractions and the pairings with one internal contraction in each cubic factor. Equivalently,

$$K = \sum_{abc} C_{abc} : x_a x_b x_c : +3\sum_c L_c x_c, \tag{10}$$

where the colons denote ordinary Gaussian normal ordering of the *couplings*. Orthogonality of these two Gaussian degrees proves (9) without subtracting large signed contributions.

Symmetry reduction. For the full complex-SYK GUE ensemble,

$$\boxed{L_c = 0, \qquad \mu_2 = 6\sum_{abc} \|C_{abc}\|^2.} \tag{11}$$

Appendix A proves this by applying the known multiplicity-free $U(N)$ operator decomposition [2]. The reduction concerns degree in Gaussian couplings, not a statement about a number of propagating bulk particles. It holds sector by sector before taking a large-N limit.

3 A complete finite-system benchmark

For $N = 4, p = 2$, a full covariance sum gives the following rational values. The table uses $w = 1$; multiply the last two columns by w^2, w^3 to restore the physical variance. Each sector trace is divided by $d_m = \binom{N}{m}$.

m	d_m	$\mathbb{E}\,\mathrm{tr}_m A$	$\mathbb{E}\,\mathrm{tr}_m A^2$	$\mathbb{E}\,\mathrm{tr}_m K^\dagger K$
0	1	0	0	0
1	4	0	0	0
2	6	12	292	0
3	4	27	942	480
4	1	36	1488	0
full	16	27/2	438	120

The only dynamical sector here is $m = 3$. At $\mathcal{J}_s = 1$ one has $w = 1/4$, hence

$$\mu_0 = \frac{219}{8}, \qquad \boxed{\mu_2 = \frac{15}{8}}. \tag{12}$$

This positive value is an ensemble moment, not a random-sample estimate. The $m = p$ block is a useful physical control: $A_p = pH_p^2$ is conserved. The full $p = 1$ model is another control, with $A = c^\dagger h^2 c$ for $H = c^\dagger h c$ and $\mu_2 = 0$.

An independent expansion of the unsymmetrized cubic expression into all 15 Wick pairings yields nine zero terms, two terms 645, and four terms $-585/2$, summing to 120. Their absolute sum is 20.5 times the answer; the positive expression (11) avoids this cancellation. The integer contraction L_c vanishes exactly in the finite calculation.

As an additional check, a three-coordinate restriction of this same matrix model breaks the full ensemble symmetry. Both Gaussian degrees then contribute: $\mu_2^{(3)} = 1/8$ and $\mu_2^{(1)} = 1/32$. A 64-node product Gaussian quadrature, exact through degree six, agrees with their sum to scaled error below 10^{-15}. This checks the term eliminated by symmetry as well as the surviving term.

4 The next analytic step

Existing q-Hermite and chord techniques solve substantial parts of embedded-ensemble spectral theory [3]. Here we first reduce the finite-N tensor norm to body-rank sums using the covariance eigenchannels of [2]. The target is the leading dependence on N, p at fixed p^2/N, then its small positive-λ expansion. The benchmark above fixes the normalization and signed contractions of that calculation.

A Why the linear Gaussian component vanishes

This is an application of the standard fermionic operator decomposition, not a new representation-theoretic theorem. Let $\mathcal{H}_r = \bigwedge^r \mathbb{C}^N$ and let $\mathcal{E}_m$ embed a p-particle coupling matrix into the m-particle Hamiltonian. The known decomposition [2], Section II, is

$$\mathrm{End}(\mathcal{H}_r) = \bigoplus_{s=0}^{\min(r,N-r)} \mathcal{R}_s. \tag{13}$$

Each $\mathcal{R}_s$ occurs once; different s are inequivalent. The summands are preserved by adjoint and entrywise complex conjugation. Normal-ordered embedding $\mathcal{E}_m$ commutes with both operations and is nonzero on every irrep common to source and target for $m \geq p$. For example its highest-weight rank-s monomial is multiplied by a nonzero real spectator factor proportional to $\binom{m-s}{p-s}$. For $m < p$, both H_m and A_m vanish.

For a Hermitian source matrix X, form the linear map

$$\mathcal{F}_m(X) = \mathbb{E}\left[K_m(G)\ \mathrm{Tr}_{\mathcal{H}_p}(GX)\right]. \tag{14}$$

The GUE measure and the sum over fermion labels in A are $U(N)$ covariant. Thus $\mathcal{F}_m$ is an equivariant real-linear map on Hermitian matrices, with a complex-linear extension to their full matrix space. Multiplicity freedom implies that on each common irrep

$$\mathcal{F}_m|_{\mathcal{R}_s} = \alpha_s \mathcal{E}_m|_{\mathcal{R}_s}. \tag{15}$$

All CAR matrices may be chosen real in the occupation basis. Conjugating G therefore conjugates K, and the GUE law is unchanged. Consequently $\mathcal{F}_m(\overline{X}) = \overline{\mathcal{F}_m(X)}$, which makes α_s real. On the other hand K_m is anti-Hermitian, whereas $\mathcal{E}_m(X)$ is Hermitian for Hermitian X; this makes α_s purely imaginary. Hence $\alpha_s = 0$. Irreps absent from the target have zero image by equivariance. We conclude $\mathcal{F}_m = 0$.

Every real Gaussian coordinate is a real linear functional of G. It follows that $\mathbb{E}[x_c K_m] = 0$. Ordinary Gaussian integration gives $\mathbb{E}[x_c K_m] = 3\sum_a (C_{aac})_m$, proving (11) in every sector. A restricted set of GUE coordinates is not invariant under $U(N)$, which explains why the independent three-coordinate test retains a nonzero linear component.

B Conventions, verification and scope

The calculation uses finite N, $1 \leq p \leq N$, the full Hermitian Gaussian covariance, $\beta = \mu = 0$, and no spectral truncation. It is ordinary Wick ordering in coupling space; it is not an identification with the q-Hermite normal ordering of the limiting Hamiltonian. For $H_c = pH_s$, $A_c = p^2 A_s$, the corresponding moments obey $\mu_{0,c} = p^4 \mu_{0,s}$ and $\mu_{2,c} = p^6 \mu_{2,s}$.

The tensor sum has not yet been reduced to a closed combinatorial expression at general N, p. Its present matrix implementation scales poorly and is a calibration tool, not the proposed large-N algorithm. A nonzero finite-N moment alone does not establish its double-scaled limit or a particle mass. Subtracting $\mathbb{E}A$ or the sample trace of A also does not remove every conserved operator. Equation (3) is insensitive to all such zero-frequency contributions.

Reproducibility. `code/singlet_wick_moments.py` constructs all coupling directions, the quadratic and cubic tensors, and their sector norms. `code/run_singlet_wick_moments.py` records full contractions for $(N, p) = (3, 1), (3, 2), (4, 2)$, the 15-pairing check, independent quadrature, and a spectral-moment comparison. Integer-valued matrix operations are checked before rational normalization; the control quadrature and diagonalization use floating point. Parameters, dependency versions, exact fractions and code hashes are stored in `moments.json` in the corresponding results directory. No disorder samples are added to the previous stage.

Literature use. We reuse the covariance decomposition of [2] and the Wick-method perspective of [3]. The latter's independent-probe final kernel does not itself evaluate the coupling-dependent A. The particle-removal moments in [4] are a further route to the adjacent-sector sums and are the next reading target. The original coupled four-point density from the dictionary note remains the all-frequency object to solve.

References

[1] S. Miyashita, Y. Sekino and L. Susskind, arXiv:2506.18054v3.

[2] V. K. B. Kota, *SU(N) Wigner–Racah algebra for the matrix of second moments of embedded Gaussian unitary ensemble of random matrices*, J. Math. Phys. **46**, 033514 (2005), arXiv:nucl-th/0410069v1.

[3] J. Tall and S. Tomsovic, arXiv:2604.14522v2, Sections 2–4 and Appendix A.

[4] V. K. B. Kota and M. Vyas, *Random Matrix Theory for Transition Strength Densities in Finite Quantum Systems: Results from Embedded Unitary Ensembles*, arXiv:1504.00759v1.